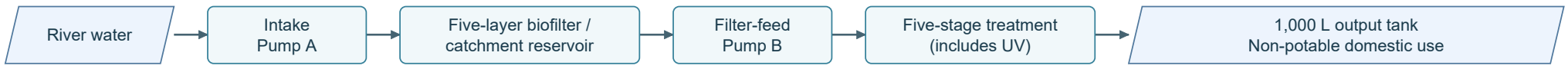
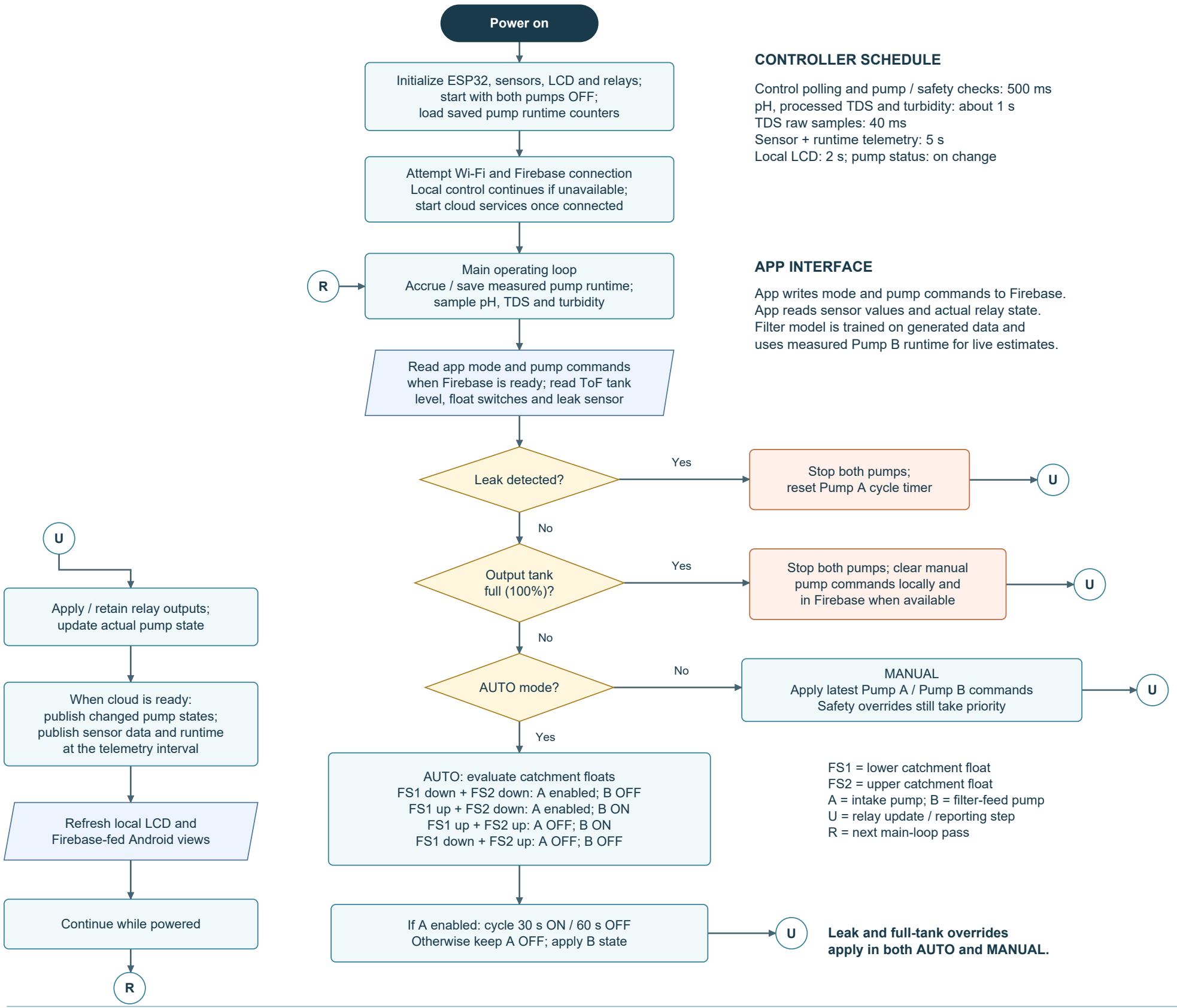
02 / HydroSense system flowchart

Water filtration and ESP32 operation | Based on thesis V23, pp. 60-62 and Appendix I | See Flowchart 01 for app navigation



Water route: biofilter media and the downstream five-stage treatment are distinct assemblies (thesis pp. 61-62).



CONTROLLER SCHEDULE

Control polling and pump / safety checks: 500 ms
pH, processed TDS and turbidity: about 1 s
TDS raw samples: 40 ms
Sensor + runtime telemetry: 5 s
Local LCD: 2 s; pump status: on change

APP INTERFACE

App writes mode and pump commands to Firebase.
App reads sensor values and actual relay state.
Filter model is trained on generated data and
uses measured Pump B runtime for live estimates.

FS1 = lower catchment float
FS2 = upper catchment float
A = intake pump; B = filter-feed pump
U = relay update / reporting step
R = next main-loop pass

Leak and full-tank overrides apply in both AUTO and MANUAL.

Timing and branch details follow Appendix I code. The prose on p. 62 differs on telemetry timing and the full-tank inequality.
Firmware uses a 5-second telemetry interval and a >=100% full-tank stop. Local safety logic continues without a cloud connection.
Sources: physical route, pp. 61-62; existing overview, p. 60; controller constants and safety / AUTO branches, pp. 166-177.